

Global Sensitivity Analysis and Quantification of Model Error for Large Eddy Simulation in Scramjet Design

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The development of scramjet engines is an important research area for advancing hypersonic and orbital flights. Progress towards optimal engine designs requires both accurate flow simulations as well as uncertainty quantification (UQ). However, performing UQ for scramjet simulations is challenging due to the large number of uncertain parameters involved and the high computational cost of flow simulations. We address these difficulties by combining UQ algorithms and numerical methods to the large eddy simulation of the HIFiRE scramjet configuration. First, global sensitivity analysis is conducted to identify influential uncertain input parameters, helping reduce the stochastic dimension of the problem and discover sparse representations. Second, as models of different fidelity are available and inevitably used in the overall UQ assessment, a framework for quantifying and propagating the uncertainty due to model error is introduced. These methods are demonstrated on a non-reacting scramjet unit problem with parameter space up to 24 dimensions, using 2D and 3D geometries with static and dynamic treatments of the turbulence subgrid model.

I. Introduction

Supersonic combustion ramjet (scramjet) engines allow propulsion systems to transition from supersonic to hypersonic flight conditions while ensuring stable combustion, potentially offering much higher efficiencies compared to traditional technologies such as rockets or turbojets. While several scramjet designs have been conceived, none to-date operate optimally.¹ This is due to difficulties in characterizing and predicting combustion properties under extreme flow conditions coupled with multiscale and multiphysics nature of the processes involved. Designing an optimal engine involves maximizing combustion efficiency while minimizing pressure losses, thermal loading, and the risk of “unstart” or flame blow-out. Achieving this, especially in the presence of uncertainty, is an extremely challenging undertaking.

An important step towards optimal scramjet design is to conduct accurate flow simulations with uncertainty quantification (UQ) in their results. While UQ in general has received substantial attention in the past decades, UQ for scramjet models is largely undeveloped, with a few exceptions.^{2,3} A comprehensive UQ study in such systems has been prohibitive due to both the large number of uncertainty sources in the predictive models, as well as the high computational cost of simulating multidimensional turbulent reacting flows. This study aims to advance the state-of-the-art on this front by developing algorithms and numerical

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methods that help set the stage for tractable UQ analysis for a realistic scramjet design. The immediate goals are to:

1. identify influential uncertain input parameters via global sensitivity analysis, which can help reduce stochastic dimension and discover sparse model representations;
2. characterize uncertainty due to model error resulted from using low-fidelity models; and
3. demonstrate these UQ methods on an initial scramjet unit problem (non-reactive, simplified geometry), and prepare extension to the full configuration.

The focal point of the current study is the HIFiRE (Hypersonic International Flight Research and Experimentation) scramjet configuration^{4,5} designed at NASA Langley Research Center, which has been the target of a mature experimental campaign with accessible data. HIFiRE is a cavity-based hydrocarbon-fueled dual-mode scramjet,⁴⁻⁷ with the entire propulsion system depicted in Figure 1(a) and the HIFiRE Direct Connect Rig (HDCR) combustor experimental setup shown in Figure 1(b). Such a dual-mode design enables transition from ramjet mode (subsonic flow in the combustor) to scramjet mode (supersonic flow in the combustor) through timing of the injectors. The combustor experiments are useful in providing data for validation as well as the design of an optimal fuel delivery schedule.

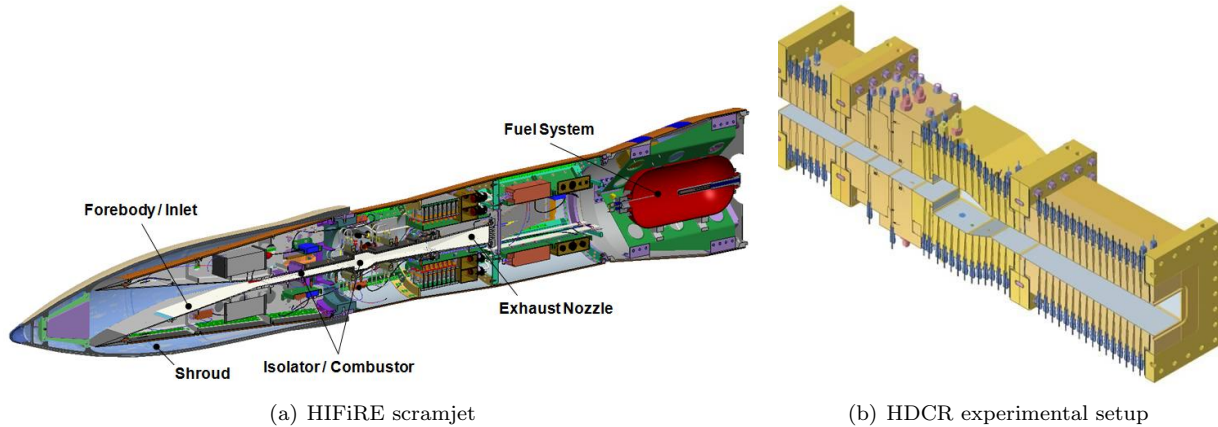


Figure 1. HIFiRE scramjet and the HDCR experimental setup.

The paper is structured as follows. Section II describes the physics and solver used for simulating the HIFiRE scramjet flow, in particular with the use of large eddy simulation techniques. We then introduce global sensitivity analysis in section III to identify the most influential input parameters of the model. In section IV, a framework is presented to capture uncertainty from model error when low-fidelity models are used. These UQ methods are then applied to the scramjet application, and the results are shown in section V. Finally, the paper ends with conclusions and future work discussions in section VI.

II. Large Eddy Simulation of the HIFiRE Scramjet

A detailed schematic of the experimental HDCR geometry is shown in Figure 2(a). The rig consists of a constant-area isolator (i.e., planar duct) attached to a combustion chamber. It includes four primary injectors that are mounted upstream of flame stabilization cavities on both the top and bottom walls. Four secondary injectors along both walls are positioned downstream of the cavities. Flow travels from left to right in the x -direction (streamwise), and the geometry is symmetric about the centerlines in both the y -direction (wall-normal) and z -direction (spanwise). Numerical simulations can take advantage of this symmetry by considering a domain that consists of only the bottom half and one set of primary and secondary injectors. Such a computational domain is highlighted by the red contours in Figure 2(a).

Even with symmetry based reductions in the size of the computational domain, the cost associated with the thousands of simulations required for the UQ analysis necessitates further simplifications. Since the current focal point is to develop, validate, and demonstrate the various UQ methodologies, a unit test problem is designed for these purposes. Calculations are performed in the region near the primary injectors

along the bottom wall ($x = 190$ to 350 mm). The domain is simplified by considering only a single primary injector and omitting the presence of the cavity. This allows a targeted investigation of the interaction between the fuel jet (JP-7 surrogate: 36% methane and 64% ethylene) and the supersonic cross flow without the effects of combustion reaction. The location of the outflow boundary is chosen to ensure that the flow is fully supersonic across the entire exit plane. The final computational domain is identified by the solid blue lines in Figure 2(b). The flow conditions of interest correspond to the freestream and fuel injection parameters reported by the HDCR experiments. Details related to LES of the full HDCR configuration are given in a companion paper by Lacaze, Vane and Oefelein.⁸

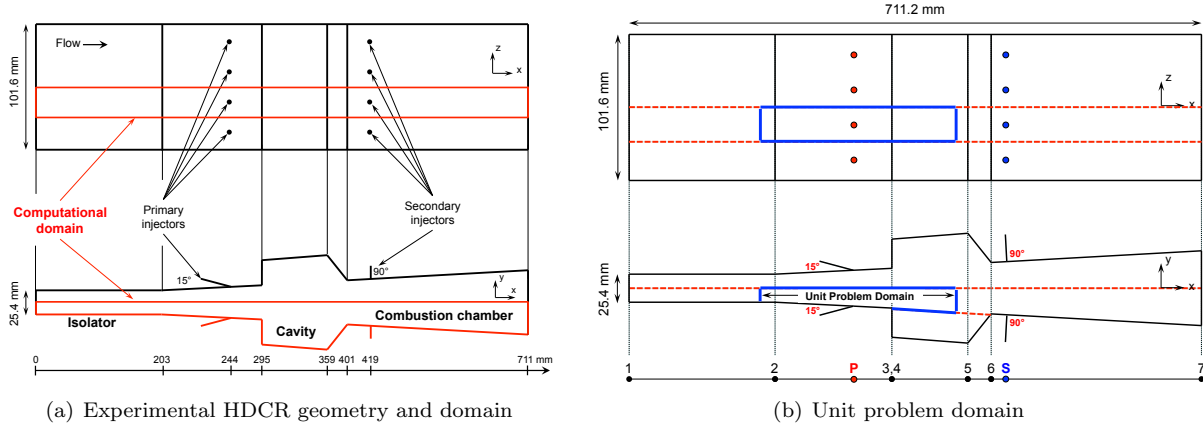


Figure 2. The schematics of the experimental HDCR geometry and its computational domain (left), and the unit problem domain (right, highlighted in solid blue lines).

II.A. LES solver: RAPTOR

LES calculations are performed using the RAPTOR code framework developed by Oefelein.^{9,10} The code is a compressible direct numerical simulation (DNS) solver that has been optimized to meet the strict algorithmic requirements imposed by the LES formalism. The theoretical framework solves the fully-coupled conservation equations of mass, momentum, total-energy, and species for a chemically reacting flow. It is designed to handle high Reynolds number, high-pressure, real-gas and/or liquid conditions over a wide Mach operating range. It also accounts for detailed thermodynamics and transport processes at the molecular level. Noteworthy is that RAPTOR is designed specifically for LES using non-dissipative, discretely conservative, staggered, finite-volume differencing. This eliminates numerical contamination of the subfilter models due to artificial dissipation and provides discrete conservation of mass, momentum, energy, and species, which is an imperative requirement for high quality LES. Representative results and case studies are given by Oefelein et al.^{11–18}

Example results for the unit problem are presented in Figure 3. Here, the instantaneous Mach number is shown across different planes along with iso-contours of ethylene (fuel component) mass fraction and Q -criterion contours that highlight coherent turbulent structures.

III. Global Sensitivity Analysis

Performing UQ analysis directly on the full physical system is very challenging, if not altogether intractable, due to the high stochastic input dimension and expensive LES computations. Often, models under different discretization levels and different assumptions are also available, and they generally trade off between computational cost and accuracy. For example, RAPTOR can perform simulations under various grid resolutions and for both 3D and 2D geometries. All models, even those with coarse grids and low fidelity, can provide *some* information about the output behavior. We thus take the following perspective: instead of selecting and using a single model for UQ analysis, we would like to combine and make use of information from different models. An intuitive interpretation of the potential advantage is that, one should extract information from inexpensive simulations when possible, and resort to the expensive (high-fidelity) simulations only for information that can only be characterized through those models. Such a multi-model

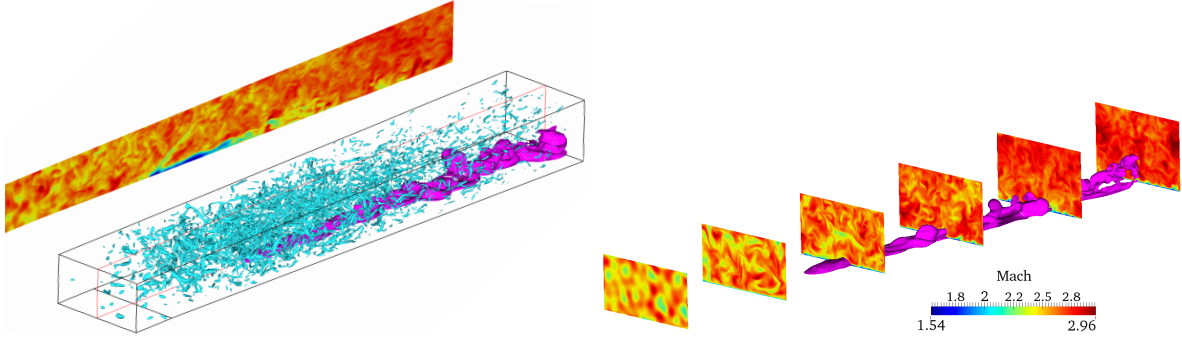


Figure 3. Typical result for the unit problem. Ethylene (fuel component): purple iso-contour ($Y_{C_2H_4} = 0.1$), turbulence: blue iso-contour (Q -criterion = 10^5 s^{-2}), and cutting planes are colored by the Mach number.

UQ analysis would be consequently much more efficient. We thus proceed under this philosophy, through the use of multilevel (ML) and multifidelity (MF) frameworks described in the next section.

While a general UQ analysis may involve many different investigations (e.g., propagation, density estimation, surrogate construction), we start by introducing global sensitivity analysis (GSA)^{19,20} in this paper. GSA can provide insights toward the uncertainty behavior of the output quantities of interest (QoIs), help identify and eliminate unimportant input parameters (e.g., reduce stochastic dimension), and provide guidance toward efficient model approximations (e.g., via sparse representation). Specifically, GSA offers a path to quantify the importance of each uncertain input parameter with respect to the predictive uncertainty of a QoI. In contrast with local sensitivity analysis, GSA reflects the overall sensitivity characteristics across the *entire* uncertain input domain.

We focus on *variance-based* properties of the input and output variables. Loosely speaking, variance of a QoI can be decomposed into contributions from the uncertainty of each input. Let λ denote the vector of all input parameters, we compute Sobol sensitivity indices²¹ to rank the components λ_i in terms of their variance contributions to that of a QoI $f(\lambda)$:

- *Main effect sensitivity* measures variance contribution solely due to the i th parameter:

$$S_i = \frac{\text{Var}_{\lambda_i} (\mathbb{E}_{\lambda_{\sim i}} [f(\lambda) | \lambda_i])}{\text{Var} (f(\lambda))}. \quad (1)$$

The notation $\lambda_{\sim i}$ refers to all components of λ *except* the i th component.

- *Joint effect sensitivity* measures variance contribution from the interaction of i th and j th parameters:

$$S_{ij} = \frac{\text{Var}_{\lambda_{ij}} (\mathbb{E}_{\lambda_{\sim ij}} [f(\lambda) | \lambda_{ij}])}{\text{Var} (f(\lambda))} - S_i - S_j. \quad (2)$$

- *Total effect sensitivity* measures variance contributions from *all* terms that involve the i th parameter:

$$S_{T_i} = \frac{\mathbb{E}_{\lambda_{\sim i}} [\text{Var}_{\lambda_i} (f(\lambda) | \lambda_i)]}{\text{Var} (f(\lambda))}. \quad (3)$$

Total effect sensitivity is particularly useful for determining which parameters have the most overall impact on the QoI and which parameters are less important. The unimportant parameters, for example, may be fixed at their nominal values without significantly affecting the output variance. Subsequently, the stochastic dimension would be reduced at a cost of only small variance approximation errors. *The primary objective of the presentation in this paper is to compute the total effect sensitivity indices.*

Traditionally, these indices can be directly estimated via various flavors of efficient Monte Carlo (MC) approaches.^{22–26} Nonetheless, the number of samples needed remains impractical for expensive models. We tackle this difficulty via two approaches. First, we take advantage of the aforementioned ML and MF formulations, to construct QoI approximations by combining simulation from models of different discretization levels (e.g., grid resolutions) and fidelity (i.e., model form). This helps transfer some of the burden from

expensive simulations to inexpensive models, and help reduce the overall sample evaluation cost. In particular, MLMF Monte Carlo (MLMF MC) is used to produce efficient sample allocation across different models. Second, we adopt polynomial chaos expansions (PCEs) to approximate the QoIs in the ML and MF forms, thereby presuming a certain degree of smoothness in the QoIs, with attendant computational savings for given accuracy requirements. Once these PCEs are available, their orthogonal polynomial basis functions allow Sobol indices to be extracted analytically from expansion coefficients without the need of additional MC sampling. We introduce MLMF and PCE in the following sections.

III.A. Multilevel and multifidelity representations

We start by first describing the multilevel concept introduced by Giles.²⁷ Consider a generic QoI produced by a model with discretization level (e.g., grid resolution) ℓ , denoted by $f_\ell(\lambda)$, where λ represents the input parameters, f_ℓ is the forward model, and $\ell = 0$ and $\ell = L$ are the coarsest and finest available resolutions, respectively. The QoI from the finest resolution can be expanded as

$$f_L(\lambda) = f_0(\lambda) + \sum_{\ell=1}^L f_{\Delta_\ell}(\lambda), \quad (4)$$

where $f_{\Delta_\ell}(\lambda) \equiv f_\ell(\lambda) - f_{\ell-1}(\lambda)$ denotes the difference terms. The intuition behind using this telescopic decomposition is that the difference terms $f_{\Delta_\ell}(\lambda)$ often tend to be better behaved and easier to characterize or approximate than the QoI directly, since some of the nonlinear behavior are expected to be subtracted out in the form of the lower-level component $f_{\ell-1}(\lambda)$. One can also view this as an exploitation of the correlation between the coarse and fine resolution evaluations. In fact, one major advantage of this setup is its ability to incorporate model evaluations across different grid resolutions. The information brought in from the inexpensive coarse simulations can help reduce the need to evaluate finer resolution models, and thus decrease the overall cost in characterizing $f_L(\lambda)$.

From here, the concept of this expansion can be utilized in various ways. For example, one may take expectations with respect to λ on each term, and generate MC samples to obtain efficient moment estimators of the fine-resolution QoI. This is known as the multilevel Monte Carlo (MLMC) method, and has extensive theoretical developments stemming from the works of Giles.²⁷ Alternatively, one may be interested to produce functional approximations of the QoI behavior:

$$f_L(\lambda) \approx \hat{f}_L(\lambda) = \hat{f}_0(\lambda) + \sum_{\ell=1}^L \hat{f}_{\Delta_\ell}(\lambda), \quad (5)$$

where $\hat{f}_\ell$ and $\hat{f}_{\Delta_\ell}$ are approximations to f_ℓ and f_{Δ_ℓ} , respectively. We take this path in this study, and adopt PCEs for these approximations (described in the next section). One motivation for this choice is that we want to leverage the observed smoothness of QoIs over the parameter space. Another reason is that the resulting PCEs provide a convenient form in which GSA may be performed, and the PCEs can also be reused in other UQ investigations of this application.

An analogous argument can be made across models of different fidelity instead of discretization levels. We refer to this other expansion as the multifidelity form. The analysis of MF approaches is more difficult and less developed than ML, since differences caused by model assumptions are more challenging to characterize systematically, and can no longer rely on properties resulting from the convergence of grid resolution. Nonetheless, MF remains a valuable tool, and has been initially explored^{28,29} with the use of sparse grids.

As will be described in detail later, we shall proceed to use a sample-based regression approach to construct the approximations $\hat{f}_0(\lambda)$ and $\hat{f}_{\Delta_\ell}(\lambda)$. The allocation of samples to be evaluated at different levels and fidelity are calculated using the MLMF MC method. This algorithm iteratively refines the sample allocations based on minimizing the variance of the MC estimators resulting from these samples, while taking into account the computational costs of different model runs. We acknowledge that this allocation procedure is not directly aimed for an optimal construction of the approximation functions, but they still provide a good general sample allocation that is useful for our study here. We refer readers to a companion paper³⁰ for an in-depth presentation of MLMF MC. Once $\hat{f}_0(\lambda)$ and $\hat{f}_{\Delta_\ell}(\lambda)$ become available, the overall approximation $\hat{f}_L(\lambda)$ can be recovered by adding them according to (5).

III.B. Polynomial chaos expansion

A polynomial chaos expansion is a spectral representation of a random variable. It provides a useful means for propagating uncertainties as an alternative to direct MC simulations of computationally expensive models. We provide a brief description of the PCE construction below, and refer readers to several books and review papers for detailed discussions.^{31–34}

With mild technical assumptions,³⁵ a real-valued random variable λ with finite variance can be expanded in the following form:

$$\lambda = \sum_{|\beta|=0}^{\infty} \lambda_{\beta} \Psi_{\beta}(\xi_1, \xi_2, \dots), \quad (6)$$

where ξ_i are i.i.d. random variables often of a simple standard form (e.g., standard Gaussian or uniform), λ_{β} are the expansion coefficients, $\beta = (\beta_1, \beta_2, \dots)$, $\forall \beta_j \in \mathbb{N}_0$, is an infinite-dimensional multi-index, $|\beta| = \beta_1 + \beta_2 + \dots$ is the ℓ_1 norm, and $\Psi_{\mathbf{i}}$ are multivariate normalized orthogonal polynomials written as products of univariate orthonormal polynomials

$$\Psi_{\beta}(\xi_1, \xi_2, \dots) = \prod_{j=1}^{\infty} \psi_{\beta_j}(\xi_j). \quad (7)$$

The PCE (6) is convergent in the mean-square sense.³⁶ The basis functions ψ_{β_j} are polynomials of order β_j in the independent variable ξ_j , orthonormal with respect to the density of ξ_j (i.e., $p(\xi_j)$):

$$\mathbb{E}[\psi_k \psi_n] = \int_{\Xi} \psi_k(\xi) \psi_n(\xi) p(\xi) d\xi = \delta_{k,n}, \quad (8)$$

where Ξ is the support of $p(\xi)$. Different choices of such ξ_j and ψ_m under the generalized Askey family are available.³⁷ We employ uniform $\xi_j \sim \mathcal{U}(-1, 1)$ and Legendre polynomials in this study. For computational purposes, the infinite sum and infinite dimension in the expansion (6) must be truncated:

$$\lambda = \sum_{\beta \in \mathcal{J}} \lambda_{\beta} \Psi_{\beta}(\xi_1, \xi_2, \dots, \xi_{n_s}), \quad (9)$$

where $\mathcal{J}$ is some index set, and n_s is some finite stochastic dimension that typically corresponds to the number of stochastic degrees of freedom in the system. For example, one popular choice for $\mathcal{J}$ is the “total-order” expansion of degree p , where $\mathcal{J} = \{\beta : |\beta| \leq p\}$. Given the expansion for the input $\lambda(\xi)$, the PCE for a QoI has the form

$$f(\lambda(\xi)) = \sum_{\beta \in \mathcal{J}} c_{\beta} \Psi_{\beta}(\xi_1, \xi_2, \dots, \xi_{n_s}). \quad (10)$$

Methods to compute its coefficients are broadly divided into two groups—intrusive and non-intrusive. The former involves substituting the expansions into the governing equations, and applying Galerkin projection to the equations, resulting in a larger and modified system for the PCE coefficients, that needs to be solved only once. The latter involves finding an approximation in the subspace spanned by the basis functions, which typically involves evaluating the original model many times. As our model is highly complicated and only available as a black-box in practice, and also for accommodating flexible choices of QoIs, we elect to take a non-intrusive route.

One such non-intrusive method relies on Galerkin projection of the solution, known as the non-intrusive spectral projection (NISP) method:

$$c_{\beta} = \mathbb{E}[f(\lambda) \Psi_{\beta}] = \int_{\Xi} f(\lambda(\xi)) \Psi_{\beta}(\xi) p(\xi) d\xi. \quad (11)$$

The integral must be estimated numerically, and thus approximately, via, for example, sparse quadrature.^{38–40} When the dimension of ξ is high, the model is expensive, and only few evaluations are available,

however, even sparse quadrature becomes impractical. In such situations, regression is a more effective method. It involves solving the following regression linear system $Gc = f$:

$$\underbrace{\begin{bmatrix} \Psi_{\beta^1}(\xi^{(1)}) & \cdots & \Psi_{\beta^N}(\xi^{(1)}) \\ \vdots & & \vdots \\ \Psi_{\beta^1}(\xi^{(M)}) & \cdots & \Psi_{\beta^N}(\xi^{(M)}) \end{bmatrix}}_G \underbrace{\begin{bmatrix} c_{\beta^1} \\ \vdots \\ c_{\beta^N} \end{bmatrix}}_c = \underbrace{\begin{bmatrix} f(\lambda(\xi^{(1)})) \\ \vdots \\ f(\lambda(\xi^{(M)})) \end{bmatrix}}_f, \quad (12)$$

where the notation Ψ_{β^n} refers to the n th basis function, c_{β^n} is the coefficient corresponding to that basis, and $\xi^{(m)}$ is the m th regression point. G is thus the regression matrix where each column corresponds to a basis and each row corresponds to a regression point. The set of M regression points is also known as the training set. We employ this regression approach for our application.

For LES, the number of affordable simulations M is expected to be drastically smaller than the number of basis terms N , leading to an extremely under-determined system that is prone to overfitting. For example, a total-order expansion of degree 3 in 24 dimensions would contain $\frac{(3+24)!}{3!24!} = 2925$ terms, while 2925 of 3D LES with $d/16$ grid resolution (i.e., each grid cell is $\frac{1}{16}$ the size of the injector diameter $d = 3.175$ mm) would take more than 64 million CPU hours! While ML and MF formulations help reduce the number expensive model simulations, we also employ compressed sensing (described in the next section) to discover sparse structure in the PCE and remove basis terms with low magnitude coefficient. Once the final PCE for the QoI is established, we can then extract the Sobol indices via the formulae:

$$S_i = \frac{1}{\text{Var}(f(\lambda))} \sum_{\beta \in \mathcal{J}_{S_i}} c_{\beta}^2, \text{ where } \mathcal{J}_i = \{\beta \in \mathcal{J} : \beta_i > 0, \beta_k = 0, k \neq i\} \quad (13)$$

$$S_{ij} = \frac{1}{\text{Var}(f(\lambda))} \sum_{\beta \in \mathcal{J}_{ij}} c_{\beta}^2, \text{ where } \mathcal{J}_{ij} = \{\beta \in \mathcal{J} : \beta_i > 0, \beta_j > 0, \beta_k = 0, k \neq i, k \neq j\} \quad (14)$$

$$S_{T_i} = \frac{1}{\text{Var}(f(\lambda))} \sum_{\beta \in \mathcal{J}_{T_i}} c_{\beta}^2, \text{ where } \mathcal{J}_{T_i} = \{\beta \in \mathcal{J} : \beta_i > 0\}, \quad (15)$$

and the QoI variance can be computed by

$$\text{Var}(f(\lambda)) = \sum_{0 \neq \beta \in \mathcal{J}} c_{\beta}^2. \quad (16)$$

III.C. Compressed sensing

Compressed sensing (CS) aims to recover the sparse solution of an under-determined linear system. Mathematically, this corresponds to minimizing the number of non-zero components of its solution vector—i.e., minimizing its ℓ_0 norm. However, ℓ_0 minimization is an NP-hard problem.⁴¹ A simpler convex relaxation approximation with ℓ_1 minimization is often used instead, and can be shown to achieve the solution of the original problem under certain conditions.⁴² This relaxed form is also known as the least absolute shrinkage and selection operator (LASSO) problem in the statistical community:^a

$$\min_c \frac{1}{2} \|Gc - f\|_2^2 \quad \text{s.t.} \quad \|c\|_1 \leq t, \quad (17)$$

where t is some threshold. An unconstrained variant (also known as the Lagrangian LASSO form) is often studied as well:

$$\min_c \frac{1}{2} \|Gc - f\|_2^2 + \eta \|c\|_1, \quad (18)$$

where η is a scalar weight. Furthermore, the use of CS for PCE construction has been previously studied, such as by Sargsyan *et al.*⁴³ with a Bayesian CS treatment combined with an iterative PCE basis enrichment

^aDifferent LASSO forms, usually with small variations such as the constant term in front of the ℓ_2 term, are also found throughout literature.

procedure. A major issue encountered, especially when data is scarce compared to the number of basis terms, is solution overfitting. This occurs when error on the training set is very different (much smaller) than the error on a separate validation set (i.e., test points), and an overfitted solution would not produce good predictions, even for interpolation. The solution would be very sensitive to the training points used, and a different training set could produce a set of entirely different sparse PCE basis terms. We will address the issue of overfitting in our setup below. More broadly, CS is a vast and active area of research across many major research fields such as signal processing, machine learning, and statistics, and many different algorithms have been developed to solve variants of the CS problem formulation; we refer interested readers to <http://dsp.rice.edu/cs> for a rich list of background papers as well as current CS developments.

We demonstrate one possible CS approach for solving the LASSO problem (other algorithms were explored as well but omitted for brevity) through the gradient projection for sparse reconstruction (GPSR).⁴⁴ GPSR targets the unconstrained LASSO variant (18) by employing a positive-negative split of the solution vector, which yields a quadratic program in its new form. A simple gradient descent with backtracking is then performed, and constraints are handled by simple projection onto the feasible space. For our numerical demonstrations, we use the MATLAB implementation GPSR 6.0 from the developers' website.

To address the issue of overfitted solutions, we return our attention to the scalar η in (18), which reflects the relative importance weight between the ℓ_1 and ℓ_2 terms; the former represents regularization and smoothing, and the latter for producing predictions that closely match the training data. A large η heavily penalizes on nonzero terms of the solution vector, forcing it toward the zero vector (underfitting); a small η heavily emphasizes the data, but may lead to solutions that are not sparse, and that *only* fit the training points but do not predict well (overfitting). A useful solution thus requires an intricate choice of η , which is a problem-dependent and nontrivial task. We examine and control the degree of overfitting by employing cross-validation (CV) to guide the choice of η . In particular, we use the K -fold CV error. The procedure involves first partitioning the full set of M training points into K equal (or approximately equal) subsets. The (normalized) K -fold CV error is given by

$$E_{CV}(\eta) \equiv \frac{1}{K} \sum_{k=1}^K \left[\frac{\sqrt{\frac{1}{M_{[k]}}} \|G_{[k]} c_{[\sim k]}(\eta) - f_{[k]}\|_2^2}{\|f_{[k]}\|_2} \right], \quad (19)$$

where $G_{[k]}$ denotes the submatrix of G that only contains rows corresponding to the k th subset, $M_{[k]}$ is the size of the k th subset, $f_{[k]}$ contains the elements of f corresponding to the k th subset, while $c_{[\sim k]}(\eta)$ is the solution to a reduced version of the unconstrained LASSO problem:

$$c_{[\sim k]}(\eta) = \underset{c}{\operatorname{argmin}} \frac{1}{2} \|G_{[\sim k]} c - f_{[\sim k]}\|_2^2 + \eta \|c\|_1, \quad (20)$$

where $G_{[\sim k]}$ denotes G but with rows corresponding the k th subset removed, and $f_{[\sim k]}$ is f with elements corresponding to the k th subset removed. The CS problem with CV treatment thus involves finding the solution c in (18) using the optimal η value recovered from

$$\eta^* = \underset{\eta}{\operatorname{argmin}} E_{CV}(\eta), \quad \text{subject to (20)}. \quad (21)$$

Solving (21) does not require the solution from the full CS system, only the K reduced systems. η^* can be found by, for example, a simple grid-search across the η -space.

IV. Embedded Representation of Model Error

Classical Bayesian model calibration typically assumes the data are consistently generated from the model—that is, the model is correct. In reality, all models are approximations to the truth, and models generally employ a tradeoff between accuracy and computational cost with various assumptions and parameterizations. For example, computational studies of turbulent combustion may rely on different geometry details, flow characteristics, turbulence assumptions, grid resolutions, and even the inclusion or removal of entire physical dimensions. As we make use of different models, it is crucial not only to acknowledge and understand—but also to develop the capability to represent, quantify, attribute, and propagate—uncertainties due to model error. We address model error, that is, error due to structural assumptions in the

model, in this study by targeting the quantities we ultimately care about in an engineering context—model predictions.

Consider two models: a high-fidelity model $g(s)$ and a low-fidelity model $f(s, \lambda)$. Both models are functions of shared continuous operating conditions s , which for simplicity consist of only relevant spatial coordinates in our LES study. The low-fidelity model also carries parameters λ which may be calibrated to provide requisite “tuning” of the model, potentially providing some compensation for its lower fidelity. Furthermore, component notations g_i and f_i shall denote the i th model observable (i.e., the categorical model output *variable*, such as T , P , etc.). We are interested in the uncertainty incurred in predictive QoIs when $f(s, \lambda)$ is used in place of $g(s)$. This entails the calibration of the low-fidelity model $f(s, \lambda)$ using data from the high-fidelity model $g(s)$.

We approach this calibration problem using a Bayesian perspective. Kennedy and O’Hagan⁴⁵ pioneered a systematic Bayesian framework for characterizing model error, by employing a linear stochastic model $q_i(s) = \rho_i f_i(s, \lambda) + \delta_i(s)$, and the calibration data $g_i(s)$ are viewed as sample realizations of $q_i(s)$. This form uses an additive Gaussian process model discrepancy term $\delta_i(s)$, and ρ_i is a constant factor. While such an approach is quite flexible in terms of attaining good fits for each calibration quantity, it poses challenges for describing predictive physical phenomena. First, the multiplicative and additive structures generally cannot guarantee the predictive quantity to maintain satisfaction of the underlying governing equations and physical laws that are instituted in f . Second, the assumed additive form makes it difficult to distinguish contributions between model error and measurement noise. Lastly, the discrepancy term, $\delta_i(s)$, is not transferable for prediction of variables outside those used for calibration (i.e., for predicting $g_l(s)$ with $l \neq i$).

We take an approach that better accommodates the aforementioned limitations while targeting more meaningful model prediction, by embedding the discrepancy term directly into the low-fidelity model such that $q_i(s) = f_i(s, \lambda + \delta_i(s))$. Furthermore, we use the following parametric stochastic model to characterize the high-fidelity data behavior:⁴⁶

$$q_i(s) = f_i(s, \lambda + \delta_i(s, \alpha_i, \xi_i)), \quad (22)$$

where α_i is a parameter of $\delta_i(\cdot)$, and ξ_i is a random variable such as, e.g., a standard normal. The uncertainty due to model error is thus encoded in both the distributions of α_i and ξ_i , where the former is reducible and can be learned from data while the latter remains fixed. In this form, the predictive quantity automatically preserves physical laws imposed in f , to the extent that this random perturbation of λ is within physical bounds. Furthermore, the contributions between the model error and data noise are separate and more readily distinguishable. Finally, while $\delta_i(\cdot)$ remains to be specific to the i th observable, we expect it to be better behaved, and generally more meaningful when extrapolated to other observables. This is supported by the fact that $\delta_i(\cdot)$ is now a correction term to the same parameter λ regardless of i , and $\delta_i(\cdot)$ would always remain in the same physical unit and likely similar magnitude. In contrast, the additive $\delta_i(s)$ from Kennedy and O’Hagan would be in entirely different physical units and potentially orders-of-magnitude different across different i . Nonetheless, extrapolation of δ_i is still needed for prediction. Finding the relationship of model error across different observables is a challenging task. We here take a simple initial approach and use a constant extrapolation, by assuming δ_i to be the same across space s , thus a random variable rather than a random process, and also across all observables i , leading to

$$q_i(s) = f_i(s, \lambda + \delta(\alpha, \xi)). \quad (23)$$

Finally, for simplifying notation, we assume s is discretized with nodes s_j , and the combined model output vector thus has the form

$$q_k = f_k(\lambda + \delta(\alpha, \xi)), \quad (24)$$

where k is the combined index of i and j . In this study, we choose to represent the now randomly perturbed parameter as a Legendre-Uniform PCE:

$$\lambda + \delta(\alpha, \xi) = \lambda + \sum_{\beta \neq 0} \alpha_\beta \Psi_\beta(\xi), \quad (25)$$

where $\Psi_\beta(\xi)$ are Legendre polynomial functions. Other PCE variants (e.g., Gauss-Hermite) are also possible; we use uniform distributions to better control the range of the perturbed parameter. When λ is multi-dimensional, the different components of $\delta(\cdot)$ may use different orders of expansions. In practice, we may

choose to embed in only certain parameter components, while keeping others at zeroth-order (i.e., treated in the classical Bayesian way). In the numerical examples of this paper, we use a simple linear expansion for the embedded λ components. A demonstration of choosing the embedding components will also be shown in the results section. The parameters are grouped together via the notation $\tilde{\alpha} \equiv (\lambda, \alpha)$. The model calibration problem thus involves finding the posterior distribution on these parameters via Bayes' theorem:

$$p(\tilde{\alpha}|D) = \frac{p(D|\tilde{\alpha})p(\tilde{\alpha})}{p(D)}, \quad (26)$$

where $D = \{g_k\}_{k=1}^K$ is the set of K calibration data points (which are equal to the high-fidelity model evaluations of the calibration QoIs in this case), $p(\tilde{\alpha})$ is the prior distribution, $p(D|\tilde{\alpha})$ is the likelihood function, and $p(D)$ is the evidence. The prior and posterior represent our states of knowledge about the uncertainty in the parameters $\tilde{\alpha}$ before and after the data set D is assimilated. To facilitate Bayesian inference and obtain the posterior in a practical manner, we will further develop the likelihood model below. Once the posterior is characterized, it can be subsequently propagated through the low-fidelity model to obtain the posterior predictive distributions for the calibration QoIs as well other model QoIs, i.e., predictions that account for both parameter and model uncertainties.

IV.A. Surrogate for low-fidelity model

Under this framework, the high-fidelity model only needs to be evaluated to generate data for Bayesian inference; this typically involves a relatively small number of evaluations. The low-fidelity model, however, needs to be run as many times as is required to perform Bayesian inference to characterize the posterior $p(\tilde{\alpha}|D)$; this entails a much larger number of evaluations in comparison. When the low-fidelity model evaluations are still expensive, a surrogate is needed. We proceed to build a surrogate (response surface) using Legendre polynomials for each QoI we plan to use for either calibration or prediction, $f_k(\cdot) \approx \hat{f}_k(\cdot)$, as functions of the overall input argument $(\lambda + \delta(\alpha, \xi))$, to replace the low-fidelity model. The approximation error between the surrogate model and the low-fidelity model is represented with a simple additive Gaussian representation, and so (24) becomes

$$q_k = f_k(\lambda + \delta(\alpha, \xi)) = \hat{f}_k(\lambda + \delta(\alpha, \xi)) + \epsilon_k, \quad (27)$$

where $\hat{f}$ is the surrogate to the low-fidelity model, and ϵ encapsulates the error of the surrogate with respect to the low-fidelity model. In this study, it is assumed $\epsilon_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{k,\text{LOO}}^2)$ and independent of the surrogate model input (but depends on k) for simplicity. The variance terms $\sigma_{k,\text{LOO}}^2$ are the leave-one-out cross-validation errors from the linear regression systems used for constructing the surrogates, and can be computed analytically and quickly (e.g., see reference⁴⁷). We emphasize that $\delta(\alpha, \xi)$ is still associated with the model discrepancy between the high-fidelity and low-fidelity models, *not* between the high-fidelity and the surrogate models.

IV.B. Likelihood approximation

We characterize the posterior $p(\tilde{\alpha}|D)$ via Markov chain Monte Carlo (MCMC) sampling, using the adaptive Metropolis algorithm.⁴⁸ The MCMC method requires the evaluation of the prior $p(\tilde{\alpha})$ and likelihood $p(D|\tilde{\alpha})$. Uniform prior distributions are adopted for simplicity, while additional choices will be explored in the future. Direct evaluation of the likelihood is intractable, since $p(D|\tilde{\alpha})$ does not have a closed form and would require either kernel density estimation (KDE) or numerical integration computations, both of which are very expensive. Furthermore, the likelihood often involves highly nonlinear and near-degenerate features (in fact, it is fully degenerate when data noise is absent⁴⁶). These challenges motivate us to seek alternatives forms that approximate the likelihood in a computationally feasible manner.

Sargsyan *et al.*⁴⁶ suggest several options based on the assumption of conditional independence between the data points. Specifically, we utilize the Gaussian approximation to the marginalized likelihood in this study:

$$p(D|\tilde{\alpha}) \approx L_G(\tilde{\alpha}) = \frac{1}{(2\pi)^{\frac{N}{2}}} \prod_{k=1}^N \frac{1}{\sigma_k(\tilde{\alpha})} \exp \left[-\frac{(\mu_i(\tilde{\alpha}) - g_k)^2}{2\sigma_k^2(\tilde{\alpha})} \right], \quad (28)$$

where

$$\mu_k(\tilde{\alpha}) = \mathbb{E} \left[\hat{f}_k(\lambda + \delta(\alpha, \xi)) + \epsilon_k \right] = \mathbb{E}_\xi \left[\hat{f}_k(\lambda + \delta(\alpha, \xi)) \right] \quad (29)$$

and

$$\sigma_k^2(\tilde{\alpha}) = \text{Var} \left[\hat{f}_k(\lambda + \delta(\alpha, \xi)) + \epsilon_k \right] = \text{Var}_\xi \left[\hat{f}_k(\lambda + \delta(\alpha, \xi)) \right] + \sigma_{k,\text{LOO}}^2 \quad (30)$$

are the mean and variance of the low-fidelity model at fixed $\tilde{\alpha} = (\lambda, \alpha)$. We estimate these moments by constructing a PCE for the outputs from propagating the PCE of the input argument in (25):

$$\hat{f}_k(\lambda + \delta(\alpha, \xi)) = \hat{f}_k \left(\lambda + \sum_{\beta \neq 0} \alpha_\beta \Psi_\beta(\xi) \right) \approx \sum_{\beta} \hat{f}_{k,\beta}(\tilde{\alpha}) \Psi_\beta(\xi). \quad (31)$$

This can be done using the NISP method described in (11) together with quadrature integration. The moments can then be computed from the expansion coefficients as

$$\mu_k(\tilde{\alpha}) \approx \hat{f}_{k,0}(\tilde{\alpha}) \quad \text{and} \quad \sigma_k^2(\tilde{\alpha}) \approx \sum_{\beta \neq 0} \hat{f}_{k,\beta}^2(\tilde{\alpha}). \quad (32)$$

When using a polynomial surrogate, and a linear input PCE is employed, NISP provides exact equality in (31).

IV.C. Posterior predictive

Once the posterior samples are generated from MCMC, one can then produce QoI predictions that account for model error by pushing the samples through (27). To simplify the presentation of this predictive distribution, we focus on its mean

$$\mathbb{E} [q_k] = \mathbb{E} \left[\hat{f}_k(\lambda + \delta(\alpha, \xi)) + \epsilon_k \right] = \mathbb{E}_{\tilde{\alpha}} \left[\hat{f}_k(\lambda + \delta(\alpha, \xi)) \right] = \mathbb{E}_{\tilde{\alpha}} [\mu_k(\tilde{\alpha})] \quad (33)$$

and variance

$$\begin{aligned} \text{Var} [q_k] &= \mathbb{E}_{\tilde{\alpha}} \left[\text{Var}_\xi \left[\hat{f}_k(\lambda + \delta(\alpha, \xi)) \right] \right] + \text{Var}_{\tilde{\alpha}} \left[\mathbb{E}_\xi \left[\hat{f}_k(\lambda + \delta(\alpha, \xi)) \right] \right] + \text{Var} [\epsilon_k] \\ &= \underbrace{\mathbb{E}_{\tilde{\alpha}} [\sigma_k^2(\lambda, \alpha)]}_{\text{model error}} + \underbrace{\text{Var}_{\tilde{\alpha}} [\mu_k(\lambda, \alpha)]}_{\text{posterior uncertainty}} + \underbrace{\sigma_{k,\text{LOO}}^2}_{\text{surrogate error}}, \end{aligned} \quad (34)$$

where all $\mathbb{E}_{\tilde{\alpha}}$ and $\text{Var}_{\tilde{\alpha}}$ are with respect to the posterior, and are computed by standard estimators using posterior samples from MCMC. The decomposition of the variance uses the law of total variance, and allows us to attribute the overall predictive variance to components due to model error, posterior uncertainty, and surrogate error. These quantities are then estimated by applying the surrogate in (31) to the MCMC samples.

V. Numerical Results

V.A. Global sensitivity analysis

We demonstrate our GSA machinery through a study with 24 input parameters shown in Table 1. Within this set, the wall temperature T_w boundary condition is a function of the continuous stream-wise coordinate x/d , and hence is a random field (RF). In this section, we will use the normalized spatial coordinates x/d , y/d , and z/d for convenience, where $d = 3.175$ mm is the diameter of the injector. T_w is thus represented using the Karhunen-Loève expansion (KLE) (see e.g., Ghanem and Spanos³¹), which is built employing the eigenstructure of the covariance function of the RF to achieve an optimal representation. We employ a Gaussian RF with a square exponential covariance structure along with a correlation length that is similar to the largest turbulent eddies (i.e., the size of the oxidizer inlet). The mean temperature profile is constructed by averaging temperature profile results from a small set of separate adiabatic simulations. The correlation

Parameter	Range	Description
Inlet boundary conditions		
p_0	[1.406, 1.554] MPa	Stagnation pressure
T_0	[1472.5, 1627.5] K	Stagnation temperature
M_0	[2.259, 2.761]	Mach number
δ_a	[2, 6] mm	Boundary layer thickness
I_i	[0, 0.05]	Turbulence intensity magnitude
L_i	[0, 8] mm	Turbulence length scale
Fuel inflow boundary conditions		
$\dot{m}_f$	[6.633, 8.107] $\times 10^{-3}$ kg/s	Mass flux
T_f	[285, 315] K	Static temperature
M_f	[0.95, 1.05]	Mach number
I_f	[0, 0.05]	Turbulence intensity magnitude
L_f	[0, 1] mm	Turbulence length scale
Turbulence model parameters		
C_R	[0.01, 0.06]	Modified Smagorinsky constant
Pr_t	[0.5, 1.7]	Turbulent Prandtl number
Sc_t	[0.5, 1.7]	Turbulent Schmidt number
Wall boundary conditions		
T_w	Expansion in 10 params of $\mathcal{N}(0, 1)$	Wall temperature represented via Karhunen-Loève expansion

Table 1. Uncertain parameters for the GSA example. The uncertain distributions are uniform across the ranges, with the exception for the wall temperature which is expressed in a KLE expansion involving 10 standard Gaussian random variables.

length employed leads to a rapid decay in characteristic-mode amplitudes, allowing us to capture about 90% of the total variance of this RF with only a 10 dimensional KLE. The wall temperature is further assumed to be constant in time.

For the ML and MF representations, we consider four combinations of grid resolutions and model fidelity: grid resolutions $d/8$ and $d/16$, for 2D and 3D simulations. A grid resolution of $d/8$ means the injector diameter $d = 3.175$ mm is discretized by 8 grid cells. Hence, $d/8$ depicts a “coarse” grid, and $d/16$ a “fine” grid. In this study, we aim to construct the following two telescopic PCEs:

$$\hat{f}_{2D,d/16}(\lambda) = \hat{f}_{2D,d/8}(\lambda) + \hat{f}_{\Delta_{2D,d/16-2D,d/8}}(\lambda) \quad (35)$$

$$\hat{f}_{3D,d/8}(\lambda) = \hat{f}_{2D,d/8}(\lambda) + \hat{f}_{\Delta_{3D,d/8-2D,d/8}}(\lambda). \quad (36)$$

The first is an exercise of ML, while the second is MF. More sophisticated PCEs representing the 3D $d/16$ model are also possible, but we do not attempt them in the current study due to the limited number of 3D $d/16$ runs.

We currently target five output observables: stagnation pressure P_{stag} , root-mean-square (RMS) of stagnation pressure $P_{stag,rms}$, Mach number M , turbulent kinetic energy TKE, and scalar dissipation rate χ . Examples of profiles for these observables across the wall-normal direction y/d , at fixed stream-wise direction $x/d = 100$, and averaged over time are shown in Figure 4 for 2D $d/8$ under various parameter settings. In the plots, the left side represents the lower wall of the chamber, with the dotted vertical line depicting the location of the wall; the right side represents the symmetry line of the chamber center. Effects of the boundary layer can be clearly seen through P_{stag} , M , and χ .

Our QoIs are the *mean* statistics of these quantities across y/d and at fixed $x/d = 100$; all QoIs are also time-averaged. For the 3D simulations, we further take the result at centerline of the span-wise direction.

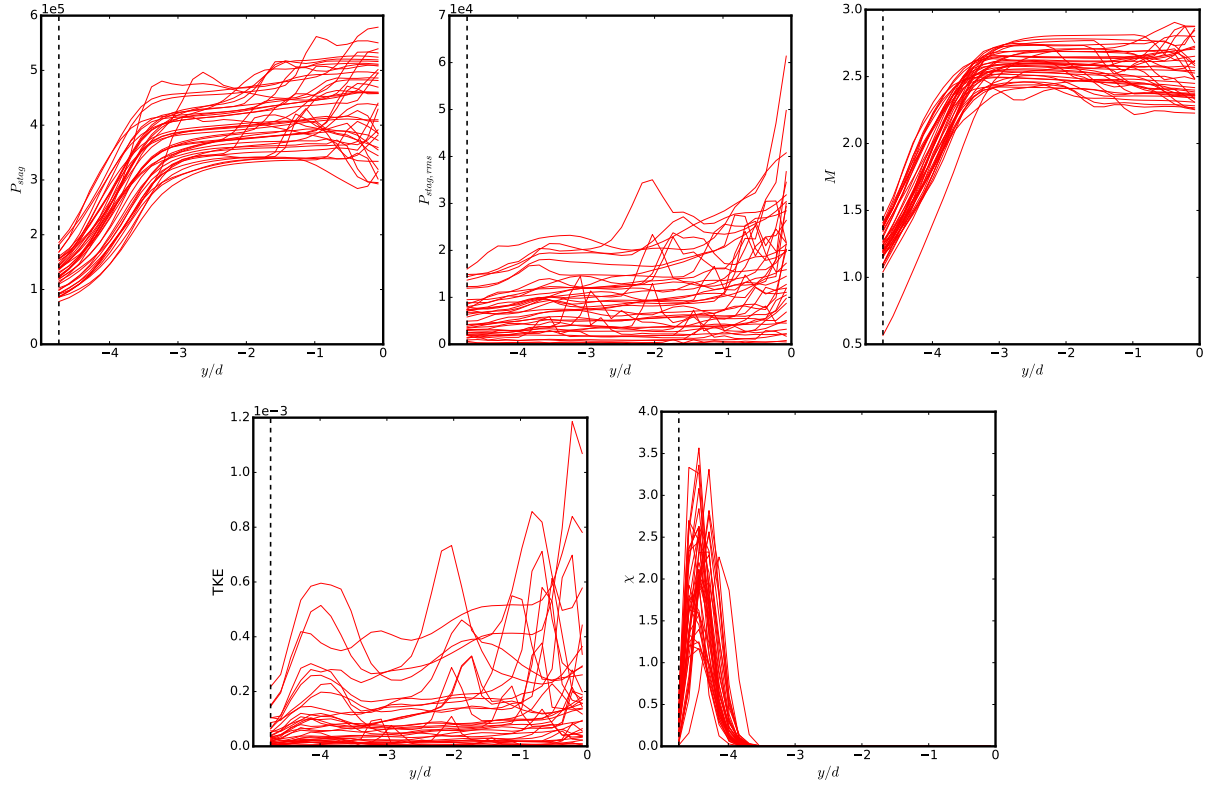


Figure 4. Profiles of five targeted observables in y/d , at fixed $x/d = 100$, and averaged over time for selected 2D $d/8$ runs of different parameter settings in the MLMF study. Each red line is the profile of an independent simulation. The vertical dotted line on the left side is the bottom wall of the chamber.

The sample allocation across different models is calculated using MLMF MC, which involves targeting to minimize the aggregate variance of MC estimators that correspond to each of the five QoIs. The total number of runs for the different models are shown in Table 2 (note even though 3D $d/16$ runs are currently not used in the PCE constructions above, they are still part of the MLMF MC allocation algorithm).

	2D	3D
$d/8$ (coarse grid)	1939	46
$d/16$ (fine grid)	79	11

Table 2. Total number of samples from the MLMF MC allocation algorithm (converged runs only). The desired general trend of low samples for expensive simulations and high samples for inexpensive simulations is evident.

The PCEs in (35) and (36) are then built for each QoI separately, using all available samples. GPSR with CV is applied to each term in those expressions to find a sparse PCE, which are then combined together before a final relative thresholding of 10^{-3} (i.e., with respect to the coefficient of largest magnitude in that expansion) is applied. For PCEs with total-order of degree 3, GPSR is able to downselect from a full set of 2925 basis terms to 187, 1336, 1676, 663, 2302 for the five QoIs in ML, and 200, 96, 308, 51, 352 for MF. The larger number of terms in ML is due to fewer available samples and a low signal-to-noise ratio for the difference between fine and coarse grid results.

The total effect sensitivity indices are plotted in Figure 5. Overall, results from both the ML and the MF expansions agree well. However, since the two expansions represent different fidelity and grid resolutions, one should not expect identical results (even under infinite samples) although it is reasonable to observe similar qualitative behavior. For both ML and MF expansions, the most sensitive inputs tend to be related to inflow conditions— M_0 , δ_a , L_i , and I_i appear to have relatively high sensitivity indices for one or more of the five QoIs. T_0 and p_0 also appear to be influential for the mean TKE, while C_R is quite important for the mean scalar dissipation rate χ . Parameters corresponding to the wall temperature KLE have generally small

contributions for most QoIs with all ten modes contributing under 5% variance. Exceptions are observed for mean $P_{stag,rms}$ receiving 25% in the ML expansion, and mean TKE receiving 67% and 21% in the ML and MF expansions, respectively. Overall, these observations are consistent with physics-based intuition, since the current unit problem does not involve combustion, and so one would expect the bulk inflow conditions to dominate the impact on the QoI behavior. The next phase of the project, involving the full HDCR geometry and with combustion enabled, is expected to shift away from these trends and reveal nonintuitive and non-obvious observations.

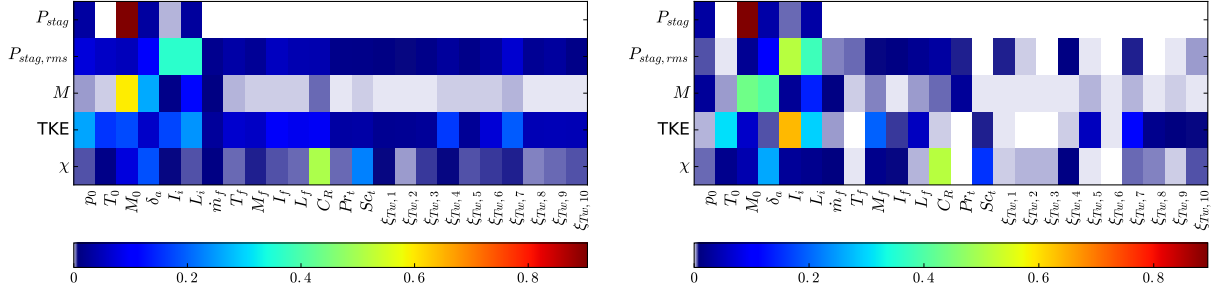


Figure 5. Total effect sensitivity indices for the ML (left) and MF (right) PCEs corresponding to (35) and (36). The rows depict the five QoIs (only the variable names are displayed in the figures, with the QoIs being the mean quantities of these variables across y/d) and the columns correspond to the 24 input parameters. $\xi_{Tw,i}$ denotes the i th KLE term for the wall temperature random field. Red spectrum indicates high sensitivity, and blue indicates low sensitivity (with grey-white being near-zero sensitivity). Sensitivity is relative within each QoI, thus one should not compare their values across different rows.

V.B. Model error

We demonstrate the embedded representation of model error via two examples. The first involves 3D LES on a grid with resolution $d/8$. The high-fidelity model uses a dynamic treatment of Smagorinsky turbulent characterization, where the Smagorinsky constant, and turbulent Prandtl and Schmidt numbers are calculated locally at every grid point. The low-fidelity model employs a static treatment, where those parameters are fixed globally across the entire grid by the user. The static version provides about 30% saving of computational time. The second example calibrates a low-fidelity 2D model from evaluations of high-fidelity 3D simulations. Both models are simulated on the $d/8$ grid, and employ the static Smagorinsky treatment. Even on this coarse grid, a 2D run requires two orders of magnitude less computational time compared to its 3D counterpart, and finer meshes would provide even greater cost savings. This is thus a realistic situation with strong tradeoff between accuracy and cost. In both cases, we would like to quantify how much model error is committed when the less expensive low-fidelity model is used.

V.B.1. Static versus dynamic Smagorinsky

For the static model, we augment the modified Smagorinsky constant $\lambda = C_R$ with an additive model error representation. It is endowed with a uniform prior $C_R \sim \mathcal{U}(0.005, 0.08)$, which is selected based on existing literature and preliminary simulation tests. The turbulent Prandtl and Schmidt numbers (Pr_t and Sc_t) are both fixed at a nominal value of 0.7. The calibration data are chosen to be the discretized profile of TKE along y/d , and at fixed location of $x/d = 100$ and span-wise centerline, for a total of 31 nodes. (All QoIs, both for calibration or prediction, are time-averaged throughout this study.) Other choices of calibration QoIs are certainly possible, and ideally we would calibrate using the same QoIs for which we are interested in making predictions. We choose TKE here for the purpose of demonstration. An interesting topic of future research is when calibration data are not available for predictive QoIs. Optimal experimental design methods can then be used to help choose from the available calibration QoIs those that are most useful for the predictive QoIs.

For our calibration, the additive term $\delta(\alpha, \xi)$ (from (27)) is set to be first-order Legendre-Uniform PCE in C_R , i.e., model error representation is embedded in C_R . The low-fidelity model surrogate, represented as a third-order Legendre polynomial, is built via regression using 9 training samples, i.e., 9 evaluations of the low-fidelity model at different C_R values. The PCEs used for the likelihood and posterior predictive are third-order Legendre-Uniform, and integration over ξ is performed using the 4-point Gaussian quadrature

rule in each dimension. MCMC is run for 10^5 samples, with 50% burn-in and thinning of every 100 samples to improve mixing.

Figure 6 depicts the static Smagorinsky model posterior predictive distributions for the profiles of TKE and stagnation pressure P_{stag} . The TKE profile constitutes the data set used for calibration, while the P_{stag} profile is extrapolative. The left column displays classical Bayesian inference results, while the right column contains results with model error representation. The black dots are the true high-fidelity evaluations, and the light grey, dark grey, and blue bands represent ± 2 standard deviations due to model error, posterior, and low-fidelity surrogate uncertainties, respectively, as broken down in (34). In this case, the data set is overall quite informative, leading to very narrow posterior distributions (dark grey band) in all figures. The classical inference results in the left column lead us to be overconfident in predictive results that do not match well with the high-fidelity model in many regions. The strength of the model error representation is evident in the right column, as the light grey bands allow much better capturing of the model-to-model discrepancy, and present a better indication of our loss in model accuracy. In this example, the model error is characterized well for the extrapolatory QoIs (P_{stag} profile, bottom-right figure). This may not always be the case, since the extrapolation of $\delta(\cdot)$ to QoIs outside those used for calibration may be inadequate, and there may be differences between high- and low-fidelity models that cannot be captured solely from parameter embedding. We will illustrate these challenges and limitations in our next example. Finally, we emphasize that all realizations generated from the posterior predictive distributions under this framework automatically satisfy the governing equations of the low-fidelity model.

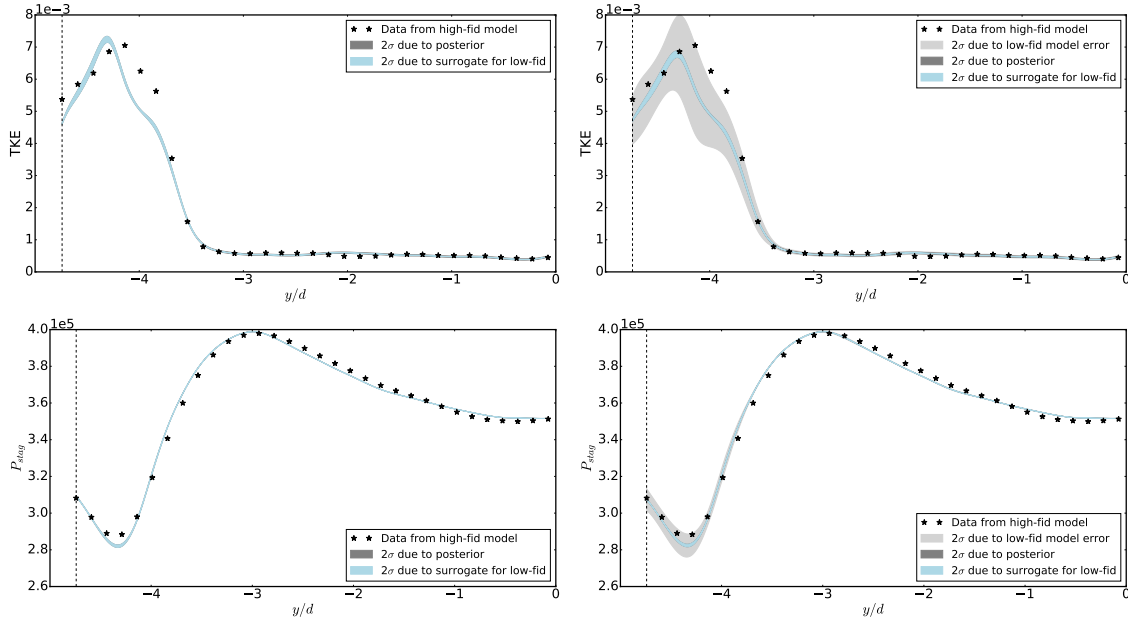


Figure 6. Posterior predictive distributions for TKE and stagnation pressure P_{stag} profiles from the static Smagorinsky model using classical Bayesian inference (left column) and model error treatment (right column). The light grey, dark grey, and blue bands correspond to contributions in (34).

V.B.2. 2D versus 3D

In the second example, we calibrate a 2D model using data from 3D model computations. The parameters for the 2D model are $\lambda = (C_R, Pr_t^{-1}, Sc_t^{-1}, I_i, I_r, L_i)$ endowed with uniform priors (see Table 3 for their definitions and prior ranges). Note that, for this example, we target the *inverse* Prandtl and Schmidt numbers instead of the non-inverted version in previous cases. The 3D calibration data are generated at a fixed condition of $\lambda_{3D}^* = (0.0297, 1/0.703, 1/0.703, 0.05, 1.0, 0.00423)$. The calibration data are chosen to be the discretized profile of scalar dissipation rate χ along y/d , at fixed $x/d = 88$ and span-wise centerline, for a total of 31 nodes.

We embed the model error representation in C_R and Sc_t^{-1} , allowing $\delta(\cdot)$ to be first-order Legendre-Uniform PCE for these two parameters only, while all other parameters are treated in the classical Bayesian

inference sense. We also enforce a triangular structure for the multivariate expansion of (25), which becomes

$$\begin{bmatrix} C_R \\ Sc_t^{-1} \end{bmatrix} + \begin{bmatrix} \alpha_{(1)}\xi_1 \\ \alpha_{(1,0)}\xi_1 + \alpha_{(0,1)}\xi_2 \end{bmatrix} \quad (37)$$

in accordance to the notation in (9), and where we have substituted the first-order Legendre-Uniform polynomial basis $\psi(\xi) = \xi$. Priors with positive support are also prescribed for $\alpha_{(1,0)}$ and $\alpha_{(0,1)}$, which then guarantee a unique distribution for each realization of the triple $(\alpha_{(1)}, \alpha_{(1,0)}, \alpha_{(0,1)})$. The decision of where to embed is guided by an initial GSA performed on the calibration QoIs: the χ profile. This is done by using the low-fidelity surrogate (27) to estimate the total sensitivity indices via the methodology in (3) for each of the spatially-discretized χ grid point. Shown in Figure 7, while sensitivity varies over y/d , the overall most sensitive parameters, especially near the bottom wall (left side of the plot) where χ values are non-zero (e.g., see Figure 8 first row), are C_R and Sc_t^{-1} . It is thus expected for the model error representation to be most effective when embedded in these two parameters. Indeed, in separate studies (results omitted), embedding in other parameters displayed less effective capturing of QoI discrepancy between the two models. The low-fidelity model surrogate is built using third-order Legendre polynomial from 500 regression samples. The PCEs used for the likelihood and posterior predictive are third-order Legendre-Uniform, and integration over ξ is performed using the 4-point Gaussian quadrature rule in each dimension. MCMC is run for 10^5 samples, with 50% burn-in and thinning of every 100 samples to improve mixing.

Parameter	Range	Description
C_R	[0.005, 0.08]	Modified Smagorinsky constant
Pr_t^{-1}	[0.25, 2.0]	Inverse turbulent Prandtl number
Sc_t^{-1}	[0.25, 2.0]	Inverse turbulent Schmidt number
I_i	[0.025, 0.075]	Inlet turbulence intensity magnitude in horizontal direction
I_r	[0.5, 1.0]	Inlet turbulence intensity vertical to horizontal ratio
L_i	[0.0, 8.0] mm	Inlet turbulence length scale

Table 3. Uncertain parameters for the 2D model in the model error study. The uncertain distributions are uniform across the ranges. Note that the prior is chosen to be uniform in the inverse of Prandtl and Schmidt numbers in this case.

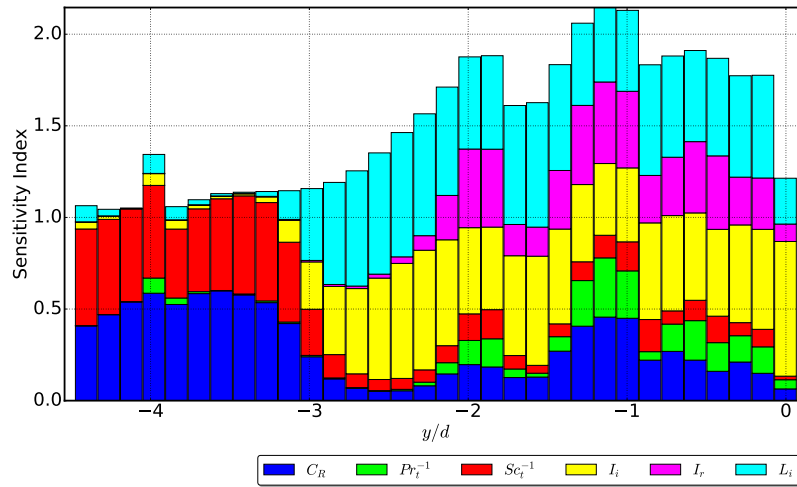


Figure 7. Total sensitivity indices for the calibration QoIs (scalar dissipation rate profile) for the 2D model in the model error study. The more sensitive parameters are good candidates to embed the model error term.

Figure 8 depicts the 2D model posterior predictive distributions for the profiles of scalar dissipation rate χ , mixture fraction Z , and Mach number M . The scalar dissipation rate profile constitutes data used for calibration, while other QoIs are extrapolated predictions. Overall, we see that the light grey band covers the model-to-model discrepancy reasonably well for χ , but there are also small regions where the high-fidelity

data are uncovered (e.g., around $y/d = -4$). For Z , we observe reasonable performance to the right of the second grid point; and for M , the light grey band is nearly non-existent (and thus has poor discrepancy coverage).

There are two important factors that explain these observed limitations of model error characterization. The first reason is a challenge for all model error approaches and not specific to the embedded representation: the extrapolation of $\delta(\cdot)$ for use on QoIs outside the calibration set (as described in (23), e.g., extrapolated from χ to Z and M profiles in this example) may be inadequate and lead to poor coverage of model-to-model discrepancy from the uncertainty bands. This is one contributing factor for the mismatched regions of the mid-right and bottom-right plots in Figure 8. However, even in some regions of the calibration QoIs (top-right plot), the light grey band does not extend to cover the high-fidelity data points. This brings us to our second reason that also demonstrates the limitation of the embedded representation: the model error bands can only be as wide as the QoI range allowed by the parameter variation (e.g., within the bounds of the uniform prior of λ). This constraint presents a difficulty when the QoI outputs from the low- and high-fidelity models are simply too different, and cannot be compensated in the low-fidelity model by varying its parameter values. In our application, the 2D model is indeed physically very different from the 3D model in many aspects, and is unable to capture many detailed physical features. (In contrast, the previous study of static and dynamic Smagorinsky models presented much closer QoI behaviors.) For instance, a bow shock structure forms in the 3D setup, and would not be portrayed in the 2D model. This difference can also change the locations where the shocks reflect, thus yielding very different “slice” profiles. Furthermore, fuel injectors in the 3D geometry are circular (not slotted), and is not equivalent to a simple extrusion of the 2D geometry. The shock strength is thus expected to be weaker in the 3D model due to the relatively smaller area of fuel injection. These insights are supported by the M profile plots, where the shocks are represented by the dips in the profiles. Additional studies (results omitted) indeed confirm that the posterior predictive bands from the right column of Figure 8 are similar to those of the prior predictive, i.e., the widest allowable by the range in Table 3. This lowered flexibility of capturing model discrepancy is the price we pay for respecting physical principles in the predictive results. At the same time, it also inspires interesting future work directions. To begin with, we would like to explore other forms of embedding that could be more advantageous, both in terms of selection of λ as well as more advanced structuring of $\delta(\cdot)$. Another possibility is to extend the framework in a hierarchy of intermediate models that offer a smoother transition between the current 3D and 2D simulations, which is also complementary to the theme of multifidelity in Section III.A. We plan to explore these techniques as we proceed to the full HDCR geometry with combustion in the next phase.

VI. Conclusions

The development of scramjet engines is an important area of research for advancing hypersonic and orbital flights. Progress towards optimal engine designs requires accurate flow simulations with uncertainty quantification (UQ) in their results. However, UQ analysis for the scramjet engine is extremely challenging due to the high number of uncertain parameters involved and the high computational cost of performing flow simulations. This paper addressed these difficulties by developing practical UQ algorithms and numerical methods and deploying them to the HIFiRE scramjet design.

We started by studying a unit problem situated in the primary injector section subdomain, and focused on the interaction between the fuel jet and the supersonic crossflow without combustion. Large eddy simulation (LES) was used to model the turbulent flow physics, and the fully coupled system of conservation laws was solved using the RAPTOR code. Global sensitivity analysis (GSA) was then conducted to identify the most influential uncertain input parameters, providing important information to help reduce the stochastic dimension and discover sparse model representations. GSA was efficiently performed by leveraging multilevel and multifidelity frameworks that can combine evaluations from different models, polynomial chaos expansion (PCE) surrogates that provide a convenient and computationally affordable form for calculating sensitivity indices, and compressed sensing that can construct these PCEs from sparsely available data. Through GSA, we were able to establish the handful of important parameters, from an initial set of 24. On a different front, we also introduced a framework for quantifying and propagating uncertainty due to model error. This technique involved embedding a correction term directly in the parameter of a model, thus allowing the preservation of physical principles in all subsequent model predictions. The correction term was represented in a stochastic and Bayesian manner, and calibrated using Markov chain Monte Carlo. Both the strengths and weaknesses of this approach were highlighted via applications of static-versus-dynamic Smagorinsky

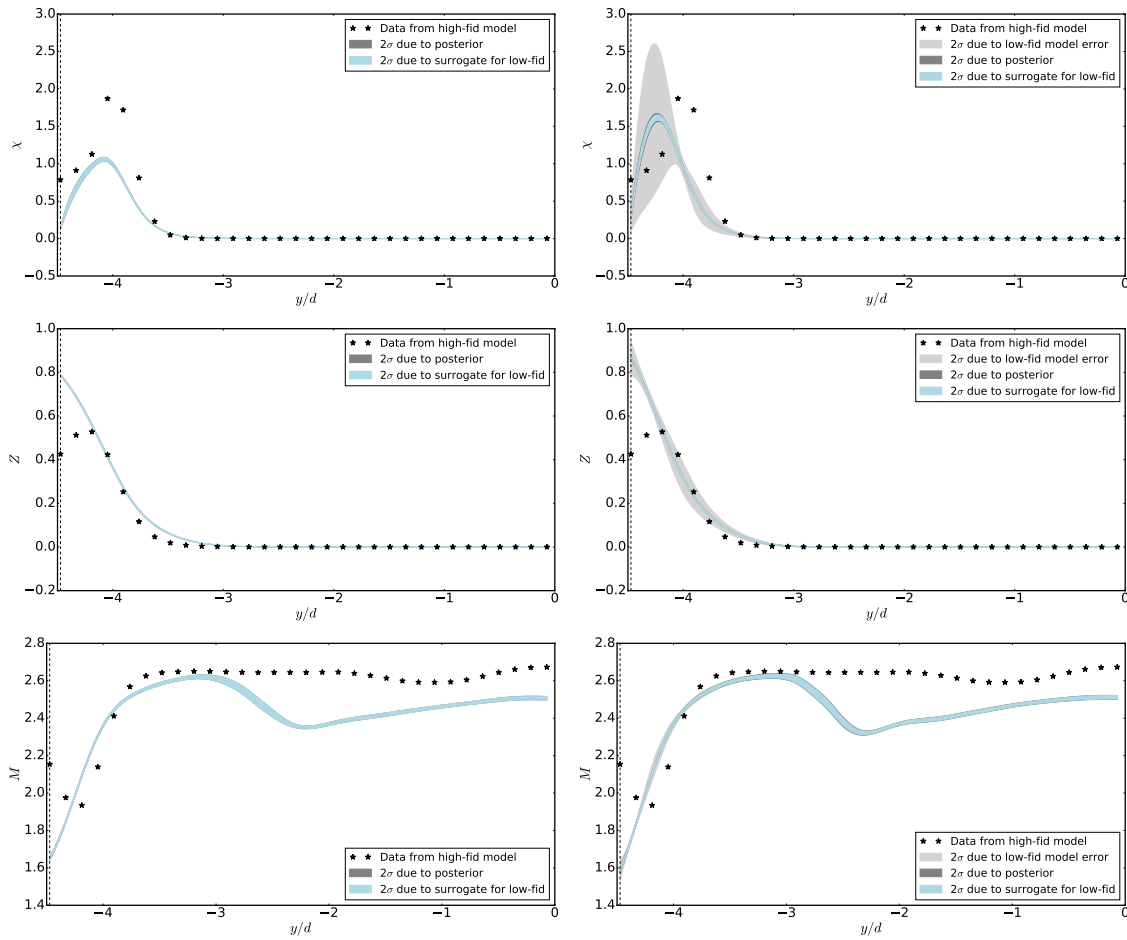


Figure 8. Posterior predictive distributions for scalar dissipation rate χ , mixture fraction Z , and Mach number M profiles from the 2D model using classical Bayesian inference (left column) and model error treatment (right column). The light grey, dark grey, and blue bands correspond to contributions in (34).

turbulent treatments, and 2D-versus-3D LES.

The logical next step is to extend these UQ techniques to the full HDCR configuration (Figure 2(a)). We expect numerous additional challenges to emerge, both for LES involving a more complex cavity geometry with combustion, and UQ analysis that will face even higher dimensional settings, increasingly expensive model evaluations, and fewer data points. Additional numerical developments will be essential to overcome these obstacles, with fruitful avenues of exploration that include adaptive and robust quadrature methods, Bayesian model selection, and efficient MCMC for model calibration.

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